Dimension

Def.: Let V be a vector space

Then $\dim(V) = \begin{cases} \text{number of elements in any } (\forall) \text{ basis of } V, & \text{if } V \text{ is f.g.} \\ \infty, & \text{if } V \text{ is not f.g.} \end{cases}$

Theorem

V: a f.g. vector space

$\Rightarrow$ Any $(\forall)$ 2 bases of V must have the same finite number of elements

Proof:

V is f.g.

$\Rightarrow$ 有 a finite basis $S = \{v_1, \cdots, v_m\}$

Suppose S' is another basis

Then $|S| \leq |S'|$ or $|S| \geq |S'|$

(需证 $|S| = |S'|$) $\leftarrow$ (假设反证: $|S| \neq |S'|$)

(Suppose on contrary) $\leftarrow$ By switching the roles of S and S' if necessary, we have $|S| < |S'|$ W.L.O.G

$\Rightarrow S' = \{u_1, \cdots, u_m, u_{m+1}, \cdots\}$

S is a basis

$$\Rightarrow \begin{cases} u_1 = \alpha_{11} v_1 + \cdots + \alpha_{m1} v_m \\ \vdots \\ u_{m+1} = \cdots \end{cases}$$

(vectors) $\leftarrow \begin{pmatrix} \alpha_{11} \\ \vdots \\ \alpha_{m1} \end{pmatrix}, \cdots, \begin{pmatrix} \alpha_{1,m+1} \\ \vdots \\ \alpha_{m,m+1} \end{pmatrix} \in \mathbb{F}^m$ 因 must be l.d.

(gamma) $\leftarrow$ $\therefore \exists \gamma_1, \gamma_2, \cdots, \gamma_{m+1}$ not all zeros,

s.t. $\begin{pmatrix} \gamma_1 \alpha_{11} + \cdots + \gamma_{m+1} \alpha_{1,m+1} \\ \vdots \\ \gamma_1 \alpha_{m1} + \cdots + \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$ $\downarrow$ (下页)

Consider $\gamma_1 \underline{u_1} + \cdots + \gamma_{m+1} u_{m+1}$

$$= \underbrace{\begin{pmatrix}\gamma_1\alpha_{11} \\ + \\ \vdots \\ + \\ \gamma_{m+1}\alpha_{1m+1}\end{pmatrix}}_{=0} \underline{v_1} + \cdots + \underbrace{\begin{pmatrix}\gamma_1\alpha_{m1} \\ + \\ \vdots \\ +\end{pmatrix}}_{=0} v_m$$

$=0$ $(\gamma_1, \cdots, \gamma_{m+1}$ not all $0)$

$\Rightarrow \{\underline{u_1}, \cdots, \underline{u_{m+1}}\}$ is l.d.

$(\subseteq \{u_1, \cdots, u_{m+1}, \cdots\} = S')$

$\Rightarrow S'$ is $\underline{\text{l.d.}}$ $\not\Leftarrow$ $\times$(l.i. $\Leftarrow$) ☆

Contradicts 不 with S' is a basis of V

Then $|S| \nless |S'|$ and $|S| \ngtr |S'|$

(前 W.L.O.G. 不)

$$\left(\left\{\boxed{E_{ij}} \;\middle|\; \begin{matrix}1\le i\le m \\ 1\le j\le n\end{matrix}\right\}\right)$$

(?)

E.g. 1. $\dim(P(\mathbb{R})) = \infty \Leftrightarrow P(\mathbb{R})$ is not f.g.

2. $P_n(\mathbb{R}) = \{p \in P(\mathbb{R}) \mid \overset{?}{\deg}(p) \le n\}$

$= \text{Span}\{\underbrace{1, x, \cdots, x^n}_{(n+1\text{个})}\}$ (有set 能 Span 它)

$\Rightarrow$

3. (见题)

Proposition 1: (Basis Extension)

V is f.g.

$\Downarrow$

it has a basis $\{u_1, \cdots, u_m\}$

Consider $B_m = \{\underbrace{v_1, \cdots, v_k}_{l.i.}, \underbrace{u_1, \cdots, u_m}_{\text{basis}}\}$

$\Downarrow$

$\text{Span}(B_m) = V$ 打卸有

还要有

① If B_m is l.i., then it is a basis of V

② If B_m is l.d.,

we have $\forall 1 \le i \le k,\ i \in \mathbb{F}$

and $\forall 1 \le j \le m,\ j \in \mathbb{F}$, α_i and β_j not all zeros

s.t. $\alpha_1 v_1 + \cdots + \alpha_k v_k + \beta_1 u_1 + \cdots + \beta_m u_m = 0$ (α_i not all zeros $\Rightarrow \exists \alpha_i \neq 0$)

(If $\forall i, \beta_i = 0 \Rightarrow \alpha_1 v_1 + \cdots + \alpha_k v_k = 0 \Rightarrow$ contradicts)

② W.L.O.G., assume $\beta_1 \neq 0$ (①)

Then $u_1 = \left(\frac{-\alpha_1}{\beta_1}\right) v_1 + \cdots$

Thus ([illegible])

Corollary : (Complementation)

(?) Proof: Let $\dim(V) = n$

Then $W \leq V \Rightarrow \dim(W) = m \leq n$

4(留 Ex.：用反证法)

Let $\{\underline{w_1}, \cdots, \underline{w_m}\}$ be a basis of W

((?) l.i. in V)

By Basis Extension,

$B = \{\underline{w_1}, \cdots, \underline{w_m}, \underline{v_1}, \cdots, \underline{v_n}\}$ is a basis of V

Let $W' = \text{Span}\{\boxed{v_{m+1}}, \cdots, \underline{v_n}\} \leq V$ (?)

Then ① $V = W + W'$

$\forall \underline{v} \in V = \text{Span}(B)$ (?)

$$(*)\quad \underbrace{\underline{v}}_{\in W+W'} = \underbrace{\alpha_1 \underline{w_1} + \cdots + \alpha_m \underline{w_m}}_{\in W} + \underbrace{\alpha_{m+1} \underline{v_{m+1}} + \cdots + \alpha_n \underline{v_n}}_{\in W'}$$

(?) ② $V = W \oplus W'$

Suppose $\underline{u} \in W \cap W'$

$\underline{u} = \cdots$ (见照)

B is l.i. $\Rightarrow (*)$ 的唯一解

是 $\beta_1 = \cdots = \beta_m = (-\gamma_{m+1}) = \cdots = (-\gamma_n) = 0$

$\Rightarrow \underline{u} = 0$ (?)

$\Rightarrow W \cap W' = \{0\}$

Linear Transformation

Def. : $T: V \to W$,
s.t. $T(\alpha \underline{v_1} + \beta \underline{v_2}) = \alpha T(\underline{v_1}) + \beta T(\underline{v_2})$

E.g.

E.g. 2 : $T = C^{\infty}(\mathbb{R}) \to C^{\infty}(\mathbb{R})$
$T(f) := f''$